

Some Useful Summation Formulae pg 16 of textbook

$$\sum_{i=1}^n 1 = n$$

$$\sum_{i=1}^n C = n * C$$

$$\sum_{i=1}^n i = \frac{n * (n+1)}{2}$$

$$\sum_{i=0}^n (n-i) = \sum_{i=0}^n i$$

$$\sum_{i=1}^n i^2 = \frac{n * (n+1) * (2n+1)}{2}$$

$$\sum_{i=0}^n 2^i = 2^{n+1} - 1$$

$$\sum_{i=0}^n C^i = \frac{C^{n+1} - 1}{C - 1}$$

$$\sum_{i=1}^n i * 2^i = (n-1) * 2^{n+1} + 2$$

$$\sum_{i=1}^n \frac{1}{i} \approx \ln n$$

$$\sum_{i=1}^n \lg i \approx n * \lg n - 1.5$$